

Introduction & Data Collection

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JupyterLab Python 3 (ipykernel)

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The purpose of this project is to analyze and compare historical stock prices of Apple (AAPL) and Tesla (TSLA) using Python.

This project applies data science techniques including:

- data collection
- data cleaning
- data visualization
- financial analysis
- machine learning

The analysis focuses on stock price trends, daily returns, volatility, moving averages, and simple price prediction using linear regression.

```
[1]: !pip install yfinance
Requirement already satisfied: yfinance in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (1.3.0)
Requirement already satisfied: pandas>=1.3.0 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (3.0.1)
Requirement already satisfied: numpy>=1.16.5 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (2.4.2)
Requirement already satisfied: requests>=2.31 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (2.32.5)
Requirement already satisfied: multitasking>=0.0.7 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (0.0.13)
Requirement already satisfied: platformdirs>=2.0.0 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (4.5.1)
Requirement already satisfied: pytz>=2022.5 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (2026.2)
Requirement already satisfied: frozendict>=2.3.4 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (2.4.7)
Requirement already satisfied: peewee>=3.16.2 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (4.0.5)
Requirement already satisfied: beautifulsoup4>=4.11.1 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (4.14.3)
Requirement already satisfied: curl_cffi>=0.15 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (0.15.0)
Requirement already satisfied: protobuf>=3.19.0 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (7.34.1)
Requirement already satisfied: websockets>=13.0 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from yfinance) (16.0)
Requirement already satisfied: soupsieve>=1.6.1 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from beautifulsoup4>=4.11.1->yfinance) (2.8.3)
Requirement already satisfied: typing-extensions>=4.0.0 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from beautifulsoup4>=4.11.1->yfinance) (4.15.0)
Requirement already satisfied: cffi>=2.0.0 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from curl_cffi>=0.15->yfinance) (2.0.0)
Requirement already satisfied: certifi>=2024.2.2 in c:\users\lenovo\appdata\local\programs\python\python314\lib\site-packages (from curl_cffi>=0.15->yfinance) (2024.2.2)
```

Project Objective:

- Analyze historical stock prices of Apple and Tesla
- Compare stock performance and volatility
- Apply Python and data science techniques
- Use simple machine learning for trend prediction

The purpose of this project is to analyze and compare Apple and Tesla stock prices using Python and financial data analysis techniques.

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JupyterLab Python 3 (ipykernel)

```
n->curl_cffi>=0.15->yfinance) (0.15.0)

[notice] A new release of pip is available: 25.3 -> 26.1
[notice] To update, run: python.exe -m pip install --upgrade pip

The following libraries are imported for data analysis, visualization, and stock market data collection.

[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import yfinance as yf

In this section, historical stock market data for Apple (AAPL) and Tesla (TSLA) are downloaded from Yahoo Finance using the yfinance library.

The dataset includes: 1.Open price 2.Close price 3.High price 4.Low price 5.Volume

[3]: apple = yf.download("AAPL", start="2022-01-01", end="2025-01-01")
tesla = yf.download("TSLA", start="2022-01-01", end="2025-01-01")

[*.....100%.....] 1 of 1 completed
[*.....100%.....] 1 of 1 completed

[4]: apple.head()

[4]:
```

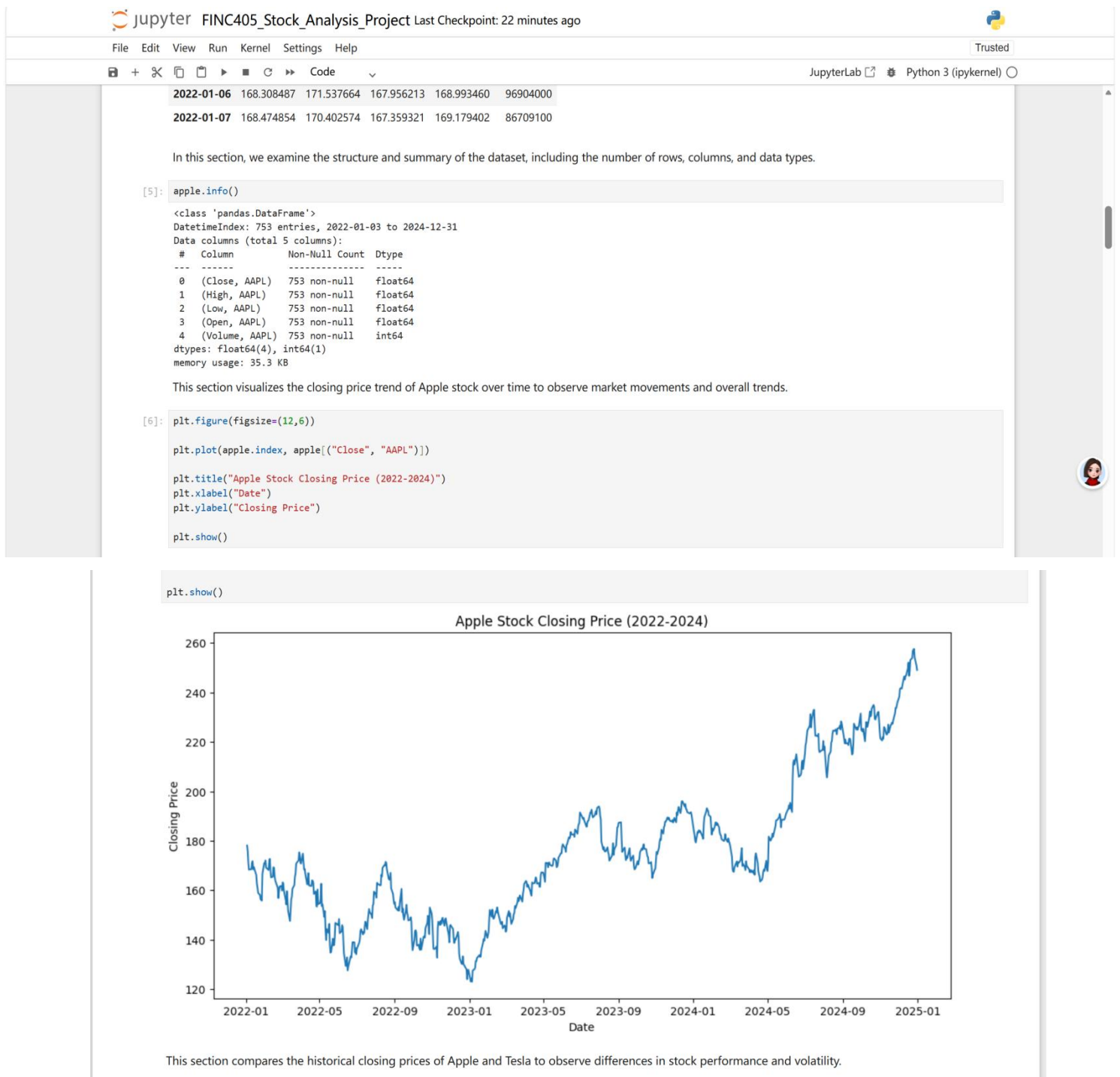
	Price	Close	High	Low	Open	Volume
Ticker	AAPL	AAPL	AAPL	AAPL	AAPL	AAPL
Date						
2022-01-03	178.103638	178.954976	173.895938	174.013357	104487900	
2022-01-04	175.843216	179.013683	175.275662	178.710339	99310400	
2022-01-05	171.165817	176.303140	170.891828	175.755162	94537600	
2022-01-06	168.308487	171.537664	167.956213	168.993460	96904000	
2022-01-07	168.474854	170.402574	167.359321	169.179402	86709100	

Pandas: data processing
NumPy: numerical calculations
Matplotlib: visualization
yfinance: stock market data collection

We used pandas and NumPy for data analysis, matplotlib for visualization, and yfinance to download stock data from Yahoo Finance.

Historical stock data for Apple and Tesla were downloaded from Yahoo Finance using the yfinance library.

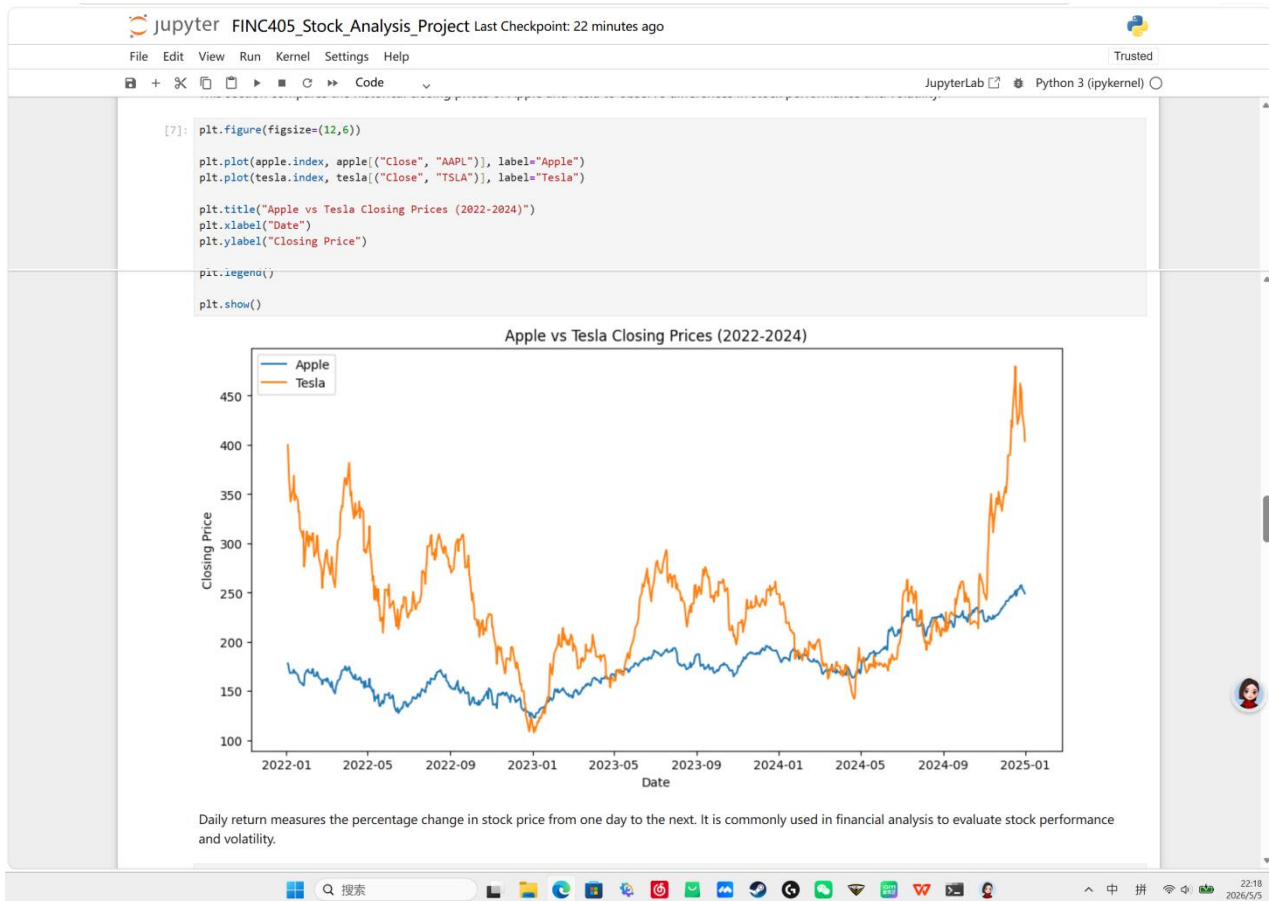
Financial Analysis & Visualization



Apple Stock Trend:

- Apple showed an overall upward trend
- Some fluctuations appeared during 2022–2023
- Stronger growth appeared in 2024

This chart shows that Apple experienced long-term growth despite short-term market fluctuations.



Apple vs Tesla:

- Tesla experienced much larger price fluctuations
- Apple showed relatively stable growth
- Tesla carried higher investment risk

Tesla showed significantly higher volatility than Apple, which indicates higher investment risk.

```
[8]: apple["Daily Return"] = apple[["Close", "AAPL"]].pct_change()
     tesla["Daily Return"] = tesla[["Close", "TSLA"]].pct_change()
```

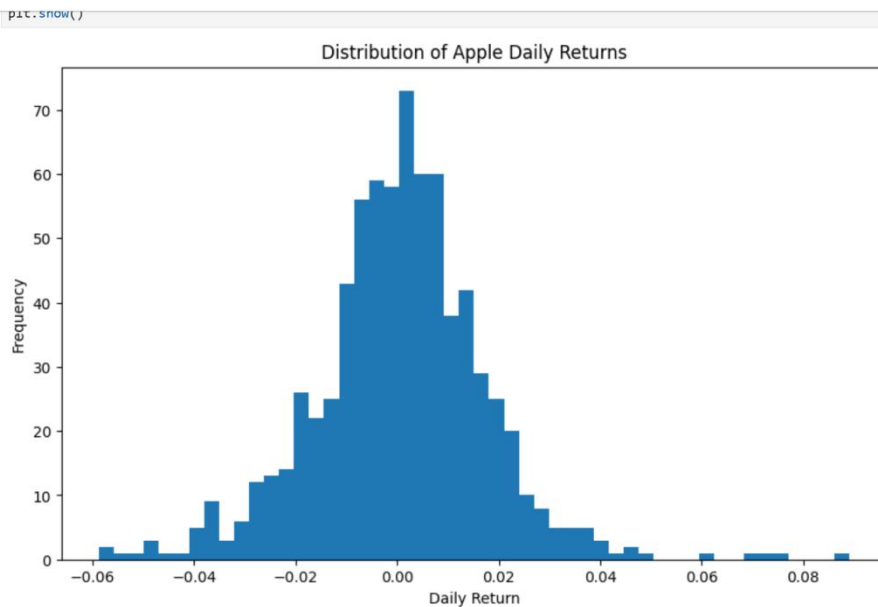
```
[9]: apple.head()
```

```
[9]:
```

	Price	Close	High	Low	Open	Volume	Daily Return
Ticker	AAPL	AAPL	AAPL	AAPL	AAPL	AAPL	
Date							
2022-01-03	178.103638	178.954976	173.895938	174.013357	104487900		NaN
2022-01-04	175.843216	179.013683	175.275662	178.710339	99310400		-0.012692
2022-01-05	171.165817	176.303140	170.891828	175.755162	94537600		-0.026600
2022-01-06	168.308487	171.537664	167.956213	168.993460	96904000		-0.016693
2022-01-07	168.474854	170.402574	167.359321	169.179402	86709100		0.000988

This section visualizes the distribution of Apple daily returns to better understand stock volatility and return behavior.

```
[10]: plt.figure(figsize=(10,6))
      plt.hist(apple["Daily Return"].dropna(), bins=50)
      plt.title("Distribution of Apple Daily Returns")
      plt.xlabel("Daily Return")
      plt.ylabel("Frequency")
      plt.show()
```



Volatility measures the degree of variation in stock returns over time. In finance, higher volatility usually indicates higher investment risk.

Daily Return Analysis:

- Daily returns were centered around zero
- Most daily changes were relatively small
- Extreme positive and negative returns occasionally occurred

Most daily returns were close to zero, but there were occasional extreme fluctuations in the market.

```
[11]: apple_volatility = apple["Daily Return"].std()
      tesla_volatility = tesla["Daily Return"].std()
      print("Apple Volatility:", apple_volatility)
      print("Tesla Volatility:", tesla_volatility)
```

```
Apple Volatility: 0.01706730574656781
Tesla Volatility: 0.03861776043609606
```

Volatility Analysis:

Apple Volatility: 0.017

Tesla Volatility: 0.038

Conclusion: Tesla is more volatile and risky.

Tesla's volatility was much higher than Apple's, which suggests greater investment risk and uncertainty.

Moving averages are commonly used in financial analysis to smooth short-term fluctuations and identify long-term market trends.

```
[12]: apple["MA_30"] = apple[("Close", "AAPL")].rolling(window=30).mean()

[13]: plt.figure(figsize=(12,6))

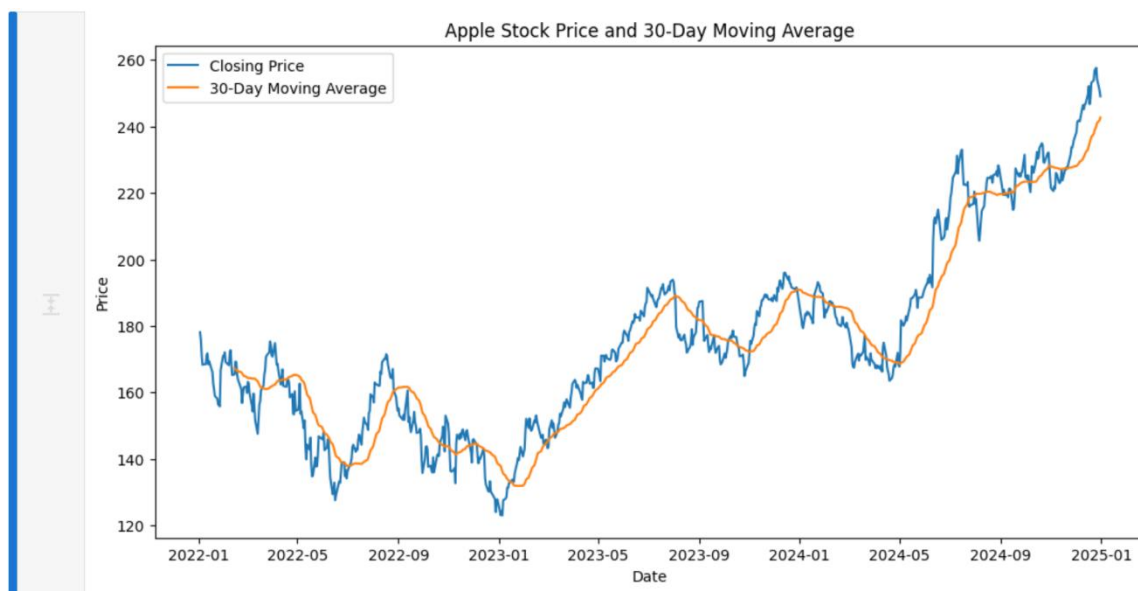
plt.plot(apple.index, apple[("Close", "AAPL")], label="Closing Price")

plt.plot(apple.index, apple["MA_30"], label="30-Day Moving Average")

plt.title("Apple Stock Price and 30-Day Moving Average")
plt.xlabel("Date")
plt.ylabel("Price")

plt.legend()

plt.show()
```



Linear regression is used to model the relationship between time and Apple stock prices. The model helps identify overall price trends and provides a simple prediction of stock movement.

30-Day Moving Average:

- Moving averages smooth short-term fluctuations
- Help identify long-term market trends
- Apple maintained long-term upward momentum

The moving average helps smooth daily price fluctuations and reveals the long-term upward trend more clearly.

Machine Learning & Conclusion

```
[14]: from sklearn.linear_model import LinearRegression
```

```
[15]: apple = apple.reset_index()
```

The dataset is prepared by converting dates into numerical values so that the regression model can analyze the relationship between time and stock prices.

```
[16]: apple["Day"] = np.arange(len(apple))
```

```
[17]: X = apple[["Day"]]
```

```
y = apple[("Close", "AAPL")]
```

The linear regression model is trained using the relationship between time and Apple stock closing prices.

```
[18]: model = LinearRegression()
```

```
model.fit(X, y)
```

```
[18]: LinearRegression
```

The trained regression model is used to generate predicted stock prices based on the historical trend of Apple stock data.

Linear Regression Model:

- Used to model the relationship between time and stock prices
- Predicted the overall trend of Apple stock prices
- Captured long-term growth direction

We used linear regression to model the relationship between time and Apple stock prices.

```
[19]: apple["Predicted Price"] = model.predict(X)
```

This visualization compares the actual Apple stock prices with the prices predicted by the linear regression model.

```
[20]: plt.figure(figsize=(12,6))
```

```
plt.plot(apple["Date"], apple[("Close", "AAPL")], label="Actual Price")
```

```
plt.plot(apple["Date"], apple["Predicted Price"], label="Predicted Price")
```

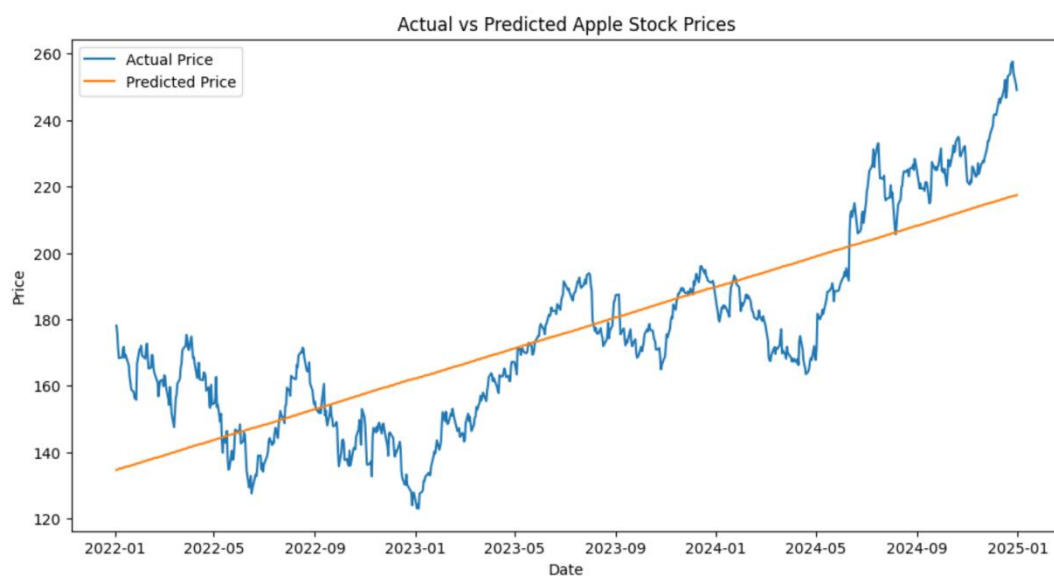
```
plt.title("Actual vs Predicted Apple Stock Prices")
```

```
plt.xlabel("Date")
```

```
plt.ylabel("Price")
```

```
plt.legend()
```

```
plt.show()
```



The R^2 score is used to evaluate how well the regression model explains variations in Apple stock prices.

Actual vs Predicted Prices:

- The model captured the overall upward trend
- Short-term fluctuations could not be fully predicted
- Stock markets contain randomness and uncertainty

The regression model captured the general upward trend, but it could not fully explain short-term market movements.

```
[21]: r_squared = model.score(X, y)
      print("R-squared:", r_squared)
      R-squared: 0.6588108504757286
```

Model Evaluation: R^2 Score = 0.658

- The model explained about 65.8% of price variation
- Performance was reasonable for a simple linear regression model

The R^2 score indicates that the model explained approximately 65.8% of the variation in Apple stock prices.

Conclusion

This project analyzed historical stock price data for Apple (AAPL) and Tesla (TSLA) using Python and data science techniques.

The analysis showed that:

- Apple demonstrated relatively stable long-term growth.
- Tesla exhibited significantly higher volatility and investment risk.
- Daily return distributions were centered around zero with occasional extreme fluctuations.
- Moving averages helped identify long-term stock price trends.
- The linear regression model successfully captured the general upward trend in Apple stock prices, although it could not fully explain short-term market fluctuations.

Overall, this project demonstrated the application of data collection, visualization, financial analysis, and machine learning techniques in stock market analysis.

Reference

1. Yahoo Finance: <https://finance.yahoo.com/>
2. yfinance Python Library: <https://pypi.org/project/yfinance/>
3. Scikit-learn Documentation: <https://scikit-learn.org/>
4. Matplotlib Documentation: <https://matplotlib.org/>
5. Pandas Documentation: <https://pandas.pydata.org/>

Conclusion:

- Apple showed relatively stable long-term growth
- Tesla exhibited significantly higher volatility
- Moving averages identified long-term trends
- Linear regression captured the general upward trend
- Python is effective for financial data analysis

Overall, this project demonstrated how Python can be used for financial analysis, visualization, and simple machine learning applications in stock market analysis.

1. Introduction

Stock market analysis is an important topic in finance because investors often evaluate stock price trends, volatility, returns, and long-term growth potential when making investment decisions. With the development of Python and data science, financial market data can now be analyzed more efficiently using programming techniques and statistical models.

This project focuses on the historical stock performance of Apple (AAPL) and Tesla (TSLA) from 2022 to 2024. Apple represents a relatively stable technology company, while Tesla is known for higher volatility and investment risk. These differences make the two companies suitable for comparison.

The main research questions of this project are:

1. How did Apple and Tesla stocks perform between 2022 and 2024?
2. Which stock showed higher volatility and investment risk?
3. Can moving averages help identify long-term stock trends?
4. Can linear regression be used to predict stock price trends?

The objective of this project is to apply Python-based data science techniques including data collection, data visualization, financial analysis, and machine learning to analyze stock market behavior.

2. Materials and Methods

The stock market data used in this project were collected from Yahoo Finance using the yfinance Python library. Historical daily stock data for Apple (AAPL) and Tesla (TSLA) were downloaded for the period from January 2022 to January 2025.

The dataset included:

- Open price
- Close price
- High price
- Low price
- Trading volume

The project was completed using Python and several libraries including:

- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- yfinance

Several analytical methods were applied in this project.

First, stock price trend analysis was conducted by visualizing the closing prices of Apple and Tesla over time in order to identify long-term market trends.

Second, daily returns were calculated using the following formula:

$$\text{Daily Return} = (P_t - P_{t-1}) / P_{t-1}$$

where P_t represents the current stock price and P_{t-1} represents the previous day's stock price.

Daily return analysis was used to measure stock performance and market fluctuations.

Third, volatility was calculated using the standard deviation of daily returns. Volatility measures the degree of price fluctuation and investment risk. Higher volatility indicates greater uncertainty and higher investment risk.

Fourth, a 30-day moving average was calculated to smooth short-term fluctuations and identify long-term stock price trends.

Finally, a linear regression model was applied to analyze the relationship between time and stock prices. The model was evaluated using the R^2 score, which measures how well the model explains stock price variation.

3. Results and Interpretation

The analysis showed that Apple stock prices demonstrated relatively stable long-term growth between 2022 and 2024. Although short-term fluctuations occurred, the overall trend remained upward.

Tesla stock prices experienced significantly larger fluctuations compared to Apple. This indicates that Tesla had higher volatility and investment risk during the sample period.

The distribution of daily returns showed that most returns were concentrated around zero, suggesting that stock prices usually changed moderately on a daily basis. However, several extreme positive and negative returns were also observed, indicating occasional periods of high market volatility.

The 30-day moving average successfully smoothed short-term fluctuations and made long-term trends easier to identify. The moving average line confirmed the long-term upward trend of Apple stock prices.

The linear regression model captured the general upward trend in Apple stock prices. The model achieved an R^2 score of approximately 0.658, indicating that the model explained about 65.8% of the variation in stock prices. However, the model could not fully explain short-term market fluctuations because stock prices are influenced by many unpredictable market factors.

4. Conclusion

This project analyzed historical stock market data for Apple and Tesla using Python and data science techniques.

The results showed that:

- Apple demonstrated relatively stable long-term growth.
- Tesla exhibited significantly higher volatility and investment risk.
- Daily return distributions were centered around zero with occasional extreme fluctuations.
- Moving averages helped identify long-term stock trends.
- Linear regression captured the general upward trend in stock prices but could not fully predict short-term market movements.

Overall, this project demonstrated the application of data collection, visualization, financial analysis, and machine learning techniques in stock market analysis.

5. References

1. Yahoo Finance: <https://finance.yahoo.com/>

2. yfinance Python Library:
<https://pypi.org/project/yfinance/>

3. Scikit-learn Documentation:
<https://scikit-learn.org/>

4. Matplotlib Documentation:
<https://matplotlib.org/>

5. Pandas Documentation:
<https://pandas.pydata.org/>